Martin Jonas A. Aranzado BSIT – 3

Capture the Flag 7: Stop Putting Everything on One Screen

Goal: Split your Student Directory into multiple screens using Flutter navigation, while using a modal for editing student information.

Instructor Note
These activities require your own problem-solving and research skills.
DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🏴 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding multiple screens.

Create:

`feature/multi-screen-navigation`

Screenshot:

- Terminal showing your current branch

```
Problems 4  Output  Debug Console  Terminal  Ports

4467440737095516l5ms. Data bytes: 468 Reply bytes: 0 Flags: 17

Application finished.
PS C:\Android Projects\IT314AB\my_first_flutter_app> git branch
* feature/conditional-ui
  feature/data-ui
  feature/exam
  feature/exam_2
  feature/student-interactions
  feature/student-list
  feature/widgets
  main
PS C:\Android Projects\IT314AB\my_first_flutter_app> git checkout -b feature/multi-screen-navigation
Switched to a new branch 'feature/multi-screen-navigation'
PS C:\Android Projects\IT314AB\my_first_flutter_app> git branch
  feature/conditional-ui
  feature/data-ui
  feature/exam
  feature/exam_2
* feature/multi-screen-navigation
  feature/student-interactions
  feature/student-list
  feature/widgets
  main
PS C:\Android Projects\IT314AB\my_first_flutter_app>
```

🚩 Flag 1 — Create the Student List Screen

Objective: Move your Student Directory into its own screen.

Create a dedicated **Student List Screen** containing your existing student list.

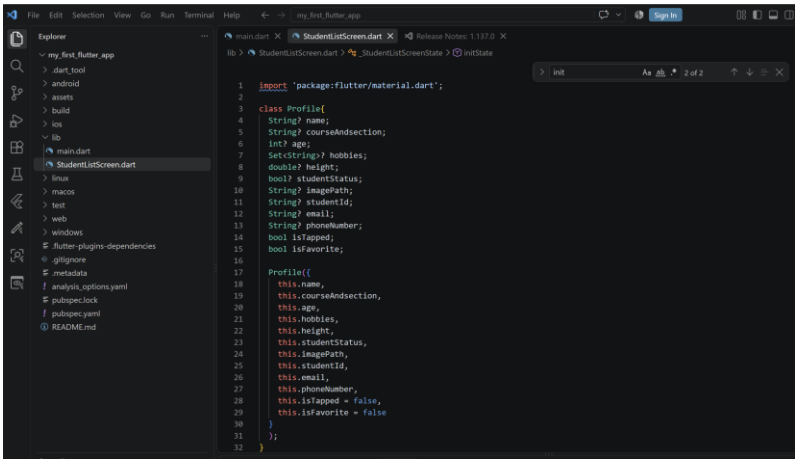
Research how Flutter creates separate screens using `Widget` classes.

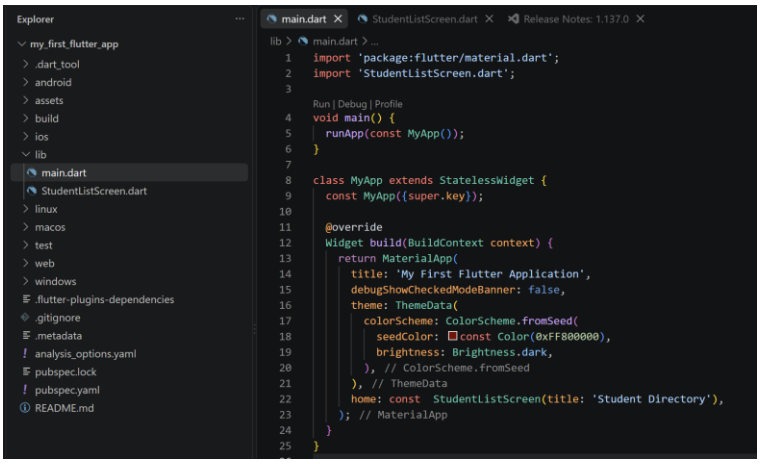
Test

The Student Directory should function as its own screen.

Screenshot:

- Student List Screen
- Student List screen code
- Application showing the screen







🏴 Flag 2 — Create the Student Details Screen

Objective: Create a separate screen for viewing a student's information.

When a student is selected, navigate:

Student List → Student Details

Research Flutter's `Navigator` and how to move between screens.

Test

1. Select a student.
2. Open the Student Details Screen. (No data displayed yet)
3. Return to the Student List using back navigation.

Screenshot:

- Student List Screen
- Student Details Screen
- Navigation code

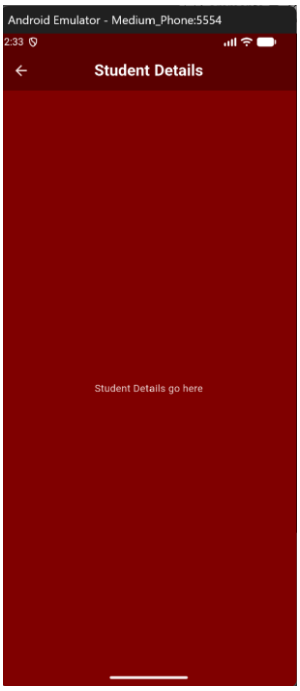
```dart
return Padding(
  padding: const EdgeInsets.only(bottom: 20.0),
  child: GestureDetector(
    onTap: (){
      Navigator.push(
        context,
          MaterialPageRoute(builder: (context) => StudentDetailsScreen(),)
      );
    },
```

```dart
import 'package:flutter/material.dart';

class StudentDetailsScreen extends StatelessWidget {
  const StudentDetailsScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: const Color(0xFF800000),
      appBar: AppBar(
        backgroundColor: const Color(0xFF5A0000),
        title: Text(
          'Student Details',
          style: const TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
          ), // TextStyle
        ), // Text
        centerTitle: true,
      ), // AppBar
      body: Center(
        child: Text('Student Details go here')) // Center
    ); // Scaffold
  }
}
```





🏴 Flag 3 — Pass Student Data

Objective: Pass the selected student's data to the Student Details Screen.

The Details Screen should display the correct information for the selected student.

Display relevant information such as:

- Name
- Student ID

- Course
- Status
- Other existing student data

Research how data can be passed between Flutter screens.

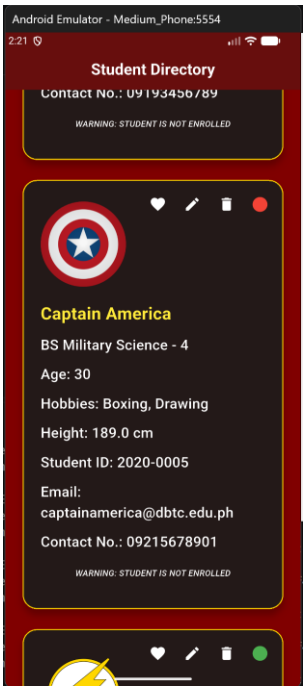
Test

Selecting different students should display their corresponding information.

Student A → Student A's Details
Student B → Student B's Details

Screenshot:

- Selected student
- Student Details
- Data-passing code






```
: ListView.builder(
    padding: const EdgeInsets.all(20.0),
      itemCount: profiles.length,
      itemBuilder: (context, index){
      final profile = profiles[index];
  return Padding(
    padding: const EdgeInsets.only(bottom: 20.0),
    child: GestureDetector(
      onTap: (){
        Navigator.push(
          context,
            MaterialPageRoute(builder: (context) => StudentDetailsScreen(profile: profile),)
        );
      },
```

🏴 Flag 4 — Edit with a Modal

Objective: Allow administrators to edit student information without creating another screen.

Add an **Edit** action to the Student Details Screen.

When selected, display an editing interface using a **modal**, such as a dialog or bottom sheet.

Research Flutter's:

- `showDialog`
- `showModalBottomSheet`

You may choose whichever approach is appropriate for your design.

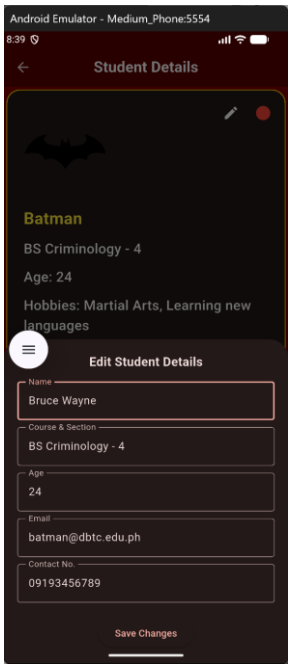
Test

1. Open Student Details.
2. Select Edit.
3. An editing modal should appear.
4. The modal should display the appropriate student's information.
5. Close or submit the modal.

Screenshot:

- Student Details Screen
- Edit modal
- Modal code





```
children: [
  IconButton(
      icon: const Icon(Icons.edit, color: Colors.white),
      onPressed: () {
        final nameController = TextEditingController(text: profile.name);
        final courseController = TextEditingController(text: profile.courseAndsection);
        final ageController = TextEditingController(text: profile.age?.toString());
        final emailController = TextEditingController(text: profile.email);
        final phoneController = TextEditingController(text: profile.phoneNumber);

  showModalBottomSheet(
    context: context,
    isScrollControlled: true,
    builder: (BuildContext context) {
      return Padding(
        padding: EdgeInsets.only(
          left: 20.0,
          right: 20.0,
          top: 20.0,
          bottom: MediaQuery.of(context).viewInsets.bottom + 20.0,
        ), // EdgeInsets.only
        child: SingleChildScrollView(
          child: Column(
            mainAxisSize: MainAxisSize.min,
            children: [
              const Text(
                'Edit Student Details',
                style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
              ), // Text
              const SizedBox(height: 15),
              TextField(
                controller: nameController,
                decoration: const InputDecoration(
                  labelText: 'Name',
                  border: OutlineInputBorder(),
                ), // InputDecoration
              ), // TextField
              const SizedBox(height: 10),
              TextField(
                controller: courseController,
                decoration: const InputDecoration(
                  labelText: 'Course & Section',
                  border: OutlineInputBorder(),
                ), // InputDecoration
              ), // TextField
              const SizedBox(height: 10),
              TextField(
                controller: ageController,
                keyboardType: TextInputType.number,
                decoration: const InputDecoration(
                  labelText: 'Age',
                  border: OutlineInputBorder(),
                ), // InputDecoration
              ), // TextField
              const SizedBox(height: 10),
              TextField(
                controller: emailController,
                decoration: const InputDecoration(
                  labelText: 'Email',
                  border: OutlineInputBorder(),
                ), // InputDecoration
              ), // TextField
              const SizedBox(height: 10),
              TextField(
                controller: phoneController,
                decoration: const InputDecoration(
                  labelText: 'Contact No.',
                  border: OutlineInputBorder(),
                ), // InputDecoration
              ), // TextField
              const SizedBox(height: 20),
              ElevatedButton(
                onPressed: () {
                  Navigator.pop(context);
                },
                child: const Text('Save Changes'),
              ), // ElevatedButton
                ],
                ), // Column
                ), // SingleChildScrollView
                ); // Padding
              },
          );
        print('Edit pressed for ${profile.name}');
      },
  ), // IconButton
  SizedBox(width: 10),
```

🚩 Flag 5 — Create the Add Student Screen

Objective: Create a separate screen for adding a new student.

Add an action from the Student List that navigates to:

Student List → Add Student

Create the **Add Student Screen**.

For now, focus on creating the screen and navigation structure.

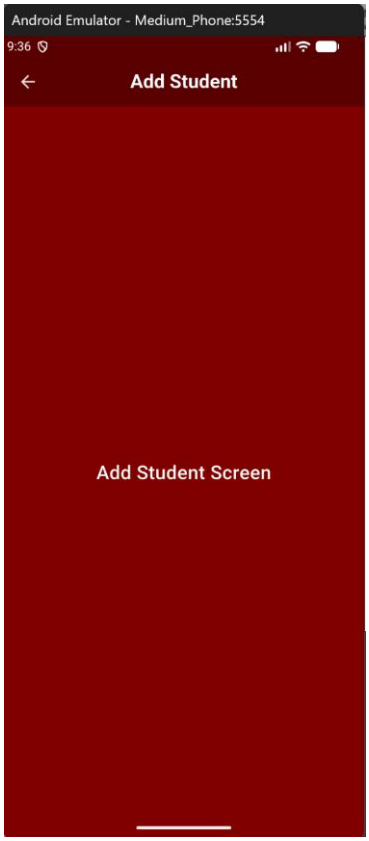
Test

Selecting Add Student should open the Add Student Screen.

Screenshot:

- Student List with Add action
- Add Student Screen
- Navigation code






```
  actions: [
  IconButton(
    icon: const Icon(Icons.add, color: Colors.white),
    onPressed: () {
      Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => AddStudentScreen(),
        ), // MaterialPageRoute
      );
    },
  ), // IconButton
],
    centerTitle: true,
  ), // AppBar
```

🚩 Flag 6 — Prepare the Add Student Screen

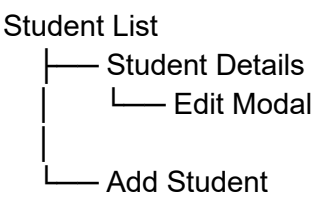
Objective: Prepare the Add Student Screen for future functionality.

Organize the screen so it is ready to receive student input.

Do not implement the student input fields yet.

The screen should simply exist as a separate destination.

Your application should now have:



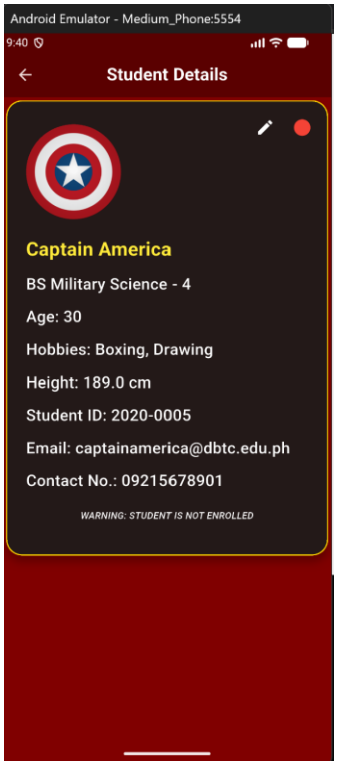
Test

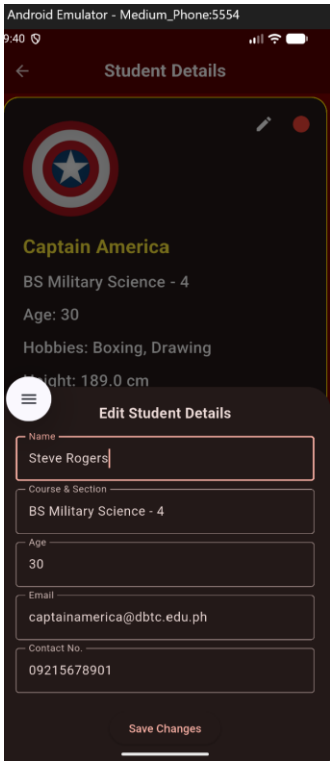
You should be able to navigate between the appropriate screens without placing everything on one screen.

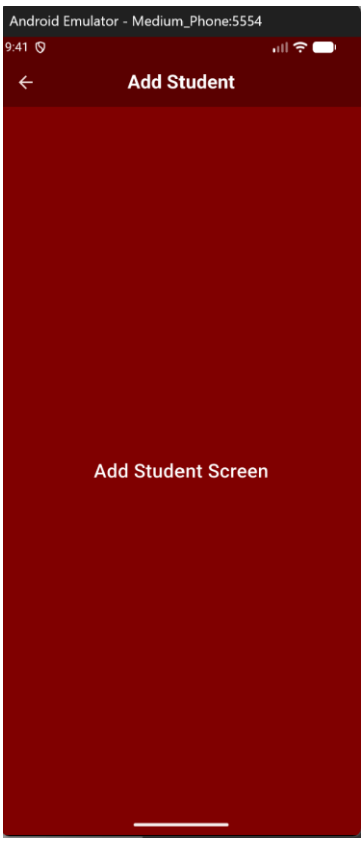
Screenshot:

- Student List
- Student Details
- Edit Modal
- Add Student Screen


Android Emulator - Medium_Phone:5554
9:39
Student Directory
Captain America
BS Military Science - 4
Age: 30
Hobbies: Boxing, Drawing
Height: 189.0 cm
Student ID: 2020-0005
Email: captainamerica@dbtc.edu.ph
Contact No.: 09215678901
WARNING: STUDENT IS NOT ENROLLED


Android Emulator - Medium_Phone:5554
9:40
Student Details
Captain America
BS Military Science - 4
Age: 30
Hobbies: Boxing, Drawing
Height: 189.0 cm
Student ID: 2020-0005
Email: captainamerica@dbtc.edu.ph
Contact No.: 09215678901
WARNING: STUDENT IS NOT ENROLLED


Android Emulator - Medium_Phone:5554
9:40
Student Details
Captain America
BS Military Science - 4
Age: 30
Hobbies: Boxing, Drawing
Edit Student Details
Name
Steve Rogers
Course & Section
BS Military Science - 4
Age
30
Email
captainamerica@dbtc.edu.ph
Contact No.
09215678901
Save Changes


Android Emulator - Medium_Phone:5554
9:41
Add Student
Add Student Screen

🚩 Flag 7 — Connect the Routes

Objective: Organize the application's screens using Flutter routes.

Create routes for the major screens:

/student-list
/student-details
/add-student

Research:

- Named routes
- `Navigator.pushNamed`
- Passing arguments through routes

The **Edit Modal should not be a route**.

Test

Verify that:

- Student List opens correctly.
- Student Details opens correctly.
- The correct student data is passed.
- Add Student opens correctly.
- Edit opens as a modal from Student Details.

Screenshot:

- Route configuration
- Navigation using routes
- Student Details receiving student data
- Edit modal

```dart
import 'package:flutter/material.dart';
import 'screens/StudentListScreen.dart';
import 'screens/StudentDetailsScreen.dart';
import 'screens/AddStudentScreen.dart';

Run | Debug | Profile
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

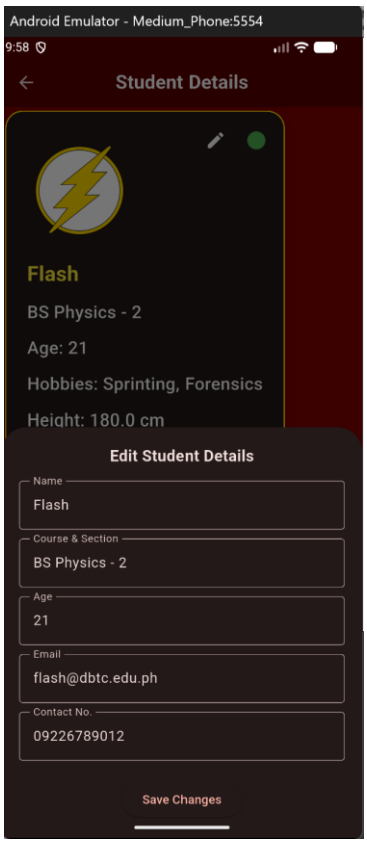
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Student Directory',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: const Color(0xFF800000),
          brightness: Brightness.dark,
        ), // ColorScheme.fromSeed
      ), // ThemeData
      initialRoute: '/student-list',

      routes: {
        '/student-list': (context) => const StudentListScreen(title: 'Student Directory'),
        '/add-student': (context) => const AddStudentScreen(),
      },

      onGenerateRoute: (settings) {
        if (settings.name == '/student-details') {
          final profile = settings.arguments as Profile;
          return MaterialPageRoute(
            builder: (context) => StudentDetailsScreen(profile: profile),
          ); // MaterialPageRoute
        }
        return null;
      },
    ); // MaterialApp
  }
}
```

```dart
child: GestureDetector(
  onTap: () {
    Navigator.pushNamed(context,'/student-details', arguments: profile,);
  },
child: Card(
```





🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Multi-Screen Student Directory.

Use a proper Git commit message.

Branch:

`feature/multi-screen-navigation`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

